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In vitro activity of eight antifungal drugs against clinical and hospital isolates of *Candida parapsilosis* complex; a multicenter study, 2023–2024

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Abstract

Background *Candida parapsilosis* complex is an emerging opportunistic pathogen associated with bloodstream infections and healthcare-associated outbreaks, particularly in intensive care units. Rising antifungal resistance underscores the need for continuous surveillance. This multicenter study assessed the in vitro susceptibility of eight antifungal agents against clinical, environmental, and healthcare worker isolates of the *C. parapsilosis* complex and investigated potential molecular mechanisms of resistance.

Methods A total of 2,600 samples were collected between May 2023 and December 2024, including 2,180 clinical specimens, 330 environmental samples, and 90 hand swabs from healthcare workers. From these, 527 *C. parapsilosis* isolates were recovered. Identification was performed using conventional and molecular approaches. Antifungal susceptibility testing followed CLSI M27 (4th edition) guidelines. Isolates exhibiting resistance were further analyzed by sequencing the ERG11 and FKS1 genes.

Results Among the 527 isolates, 510 originated from clinical settings, 6 from reusable breast pump sets, and 11 from healthcare worker hands. Overall, 10 isolates (1.9%) demonstrated resistance according to CLSI breakpoints, all of which were clinical in origin. Resistance was detected to fluconazole (0.38%), caspofungin (0.95%), and combined caspofungin–anidulafungin resistance (0.57%). The MIC values for fluconazole and caspofunginwere 4 µg/mL, indicating reduced susceptibility among clinical isolates. Voriconazole, clotrimazole, and amphotericin B exhibited strong in vitro activity. No resistance-associatedmutations were identified in ERG11 or FKS1.

Conclusions Despite the absence of detectable ERG11 or FKS1 mutations, reduced susceptibility to azoles and echinocandins was observed, primarily among clinical isolates. Environmental and healthcare worker isolates showed lower MIC values, suggesting minimal antifungal selection pressure outside patient care environments. These findings

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point to emerging antifungal resistance in *C. parapsilosis* clinical isolates in Iran and emphasize the importance of ongoing surveillance and molecular monitoring to inform therapeutic and infection control strategies. However more studies are required to correlate these findings with clinical outcomes.

Keywords *Candida parapsilosis*, Antifungal resistance, *ERG11*, *FKS1*, Nosocomial infections

Introduction

In recent years, there has been a global increase in the frequency of severe fungal infections, accompanied by a significant shift in the epidemiological landscape. While *Candida albicans* was the predominant species causing fungal infections over the past century, non-*albicans Candida* (NAC) species are now becoming more common [1]. *C. parapsilosis* is a yeast commonly found on human skin and is considered a standard component of the skin's microbiome [2]. However, it can also spread in healthcare settings, as it can be transmitted by healthcare workers'(HCWs) hands. Additionally, *C. parapsilosis* has a tendency to form biofilms on catheters and other medical devices, making it a particular concern for hospitalized patients, especially those who are immunocompromised [3]. Globally, *C. parapsilosis* accounts for approximately 16% of adult candidemia cases, making it the third most common *Candida* species causing invasive candidiasis [4]. It is particularly impactful on newborns in neonatal intensive care units, representing 33% of all invasive *Candida* infections and 80% of infections associated with NAC species in premature infants [2, 5]. Historically, *C. parapsilosis* has been considered susceptible to fluconazole, which is a standard antifungal treatment regimen for *Candida* infections [6]. The COVID-19 pandemic has significantly increased fluconazole resistance in *C. parapsilosis* [7]. The rise in resistance is likely due to several factors, including a surge in patient admissions and the widespread use of antifungal medications during the pandemic. While echinocandins are now the recommended first line treatment for invasive candidiasis, fluconazole remains a crucial part of antifungal therapy. Resistant strains of *C. parapsilosis* can thrive even when exposed to therapeutic levels of fluconazole, which contributes to treatment failures. Azole resistance in *C. parapsilosis* is most frequently linked to mutations in the *ERG11* gene. This gene encodes lanosterol demethylase, an essential enzyme for maintaining fungal cell membrane integrity and a target of azole antifungals. Mutations in *ERG11* that lead to changes in amino acid, particularly the Y132F substitution, can affect the ability of azoles to bind to and inhibit this enzyme, thereby resulting in resistance [4]. The challenges associated with *C. parapsilosis* infections are heightened by their elevated minimum inhibitory concentration (MIC) values for first line antifungal treatments, specifically echinocandins, when compared to *C. albicans* and *C. glabrata*. This highlights important differences among species [8]. Resistance to echinocandin in the *C. parapsilosis* complex is primarily linked to active mutations in the hotspot regions (HS1, HS2) of the *FKS1* gene, which encodes the enzyme complex responsible for 1,3-β-D-glucan synthase. Mutations in these *FKS1* hotspot regions have been associated with treatment failure in patients with candidemia, particularly in those who have previously been exposed to echinocandin antifungal therapy [9]. Key challenges presented by this fungal species include the urgent need for effective surveillance measures to prevent and control hospital outbreaks, thereby protecting vulnerable patient populations from direct transmission. Antifungal susceptibility testing is increasingly important for detecting the rising resistance to fluconazole in *C. parapsilosis*, which is growing at a faster rate than in other *Candida* species [10]. The lack of comprehensive data regarding the clinical outcomes, antifungal susceptibility patterns, and the underlying molecular mechanisms of antifungal resistance for *C. parapsilosis* isolates on a national level in Iran lead researchers to conduct this multicenter study, aiming to address these gaps in knowledge. Elucidating these factors will help improve the clinical management of *C. parapsilosis* infections and provide insights into the extent to which infection control strategies need to be implemented. In this observational study, we evaluated antifungal susceptibility of eight antifungal drugs, including fluconazole (FLZ), itraconazole (ITZ), voriconazole (VRZ), clotrimazole (CLO), miconazole (MZ), caspofungin (CAS), anidulafungin (AFG), and amphotericin B (AmB). This assessment was conducted on a large number of clinical, environmental, and healthcare worker derived *C. parapsilosis* isolates in Iran. Additionally, we performed molecular analysis on resistant isolates to investigate potential mutations in key resistance-associated genes, specifically *ERG11* and *FKS1*, particularly focusing on their known hotspot regions. This approach aimed to identify the underlying genetic determinants of azole and echinocandin resistance within the *C. parapsilosis* complex.

Methods

Clinical,environmental, and healthcare workers' hand sampling

This study received approval from the Ethics Committee of Mazandaran University of Medical Sciences, under the approval number IR.MAZUMS.REC.1402.365, dated September 13, 2023. All procedures were conducted according to the ethical guidelines and regulations.

A total of 2600 samples were collected, including 2180 clinical samples, 330 environmental samples, and 90 hand swabs from healthcare workers. 527 *C. parapsilosis* isolates were recovered between May 2023 and December 2024. These isolates were sourced from various contexts. including clinical, surrounding environmental, and the hand of healthcare workers' (HCWs). Although isolates obtained from healthcare workers were analyzed and reported separately, they are considered part of the hospital environment for epidemiological purposes. Clinical samples were obtained from various centers in Iran through body swabs, including the axillae, groin, ears, mouth, rectum, wound, and tissue specimens from burn sites. Additionally, blood samples, scalp, nails, neonatal skin, bronchial aspirates, sputum, catheter sites, and diabetic foot ulcers were included in the study. Each clinical isolate represented a single patient, and duplicate isolates from the same individual were excluded. A total of 330 environmental and 90 hand samples from HCWs' were collected from five educational hospitals affiliated with Mazandaran University of Medical Sciences. Environmental samples were taken from several areas of the hospital, including inpatient wards, intensive care units (ICUs), neonatal units, outpatient clinics, and patient rooms. All samples were collected using sterile swabs placed in sterile transport media from inanimate surfaces such as bed rails, cardiac monitors, ventilators, infusion pumps, dialysis devices, sinks, cabinets near patient beds, and doorknobs. The samples were transported immediately to the laboratory for microbiological analysis to identify potential pathogenic organisms and conduct susceptibility testing. In addition to clinical isolates collected from the five mentioned hospitals, samples were also obtained from other referral centers. Hand swabs were taken from 90 active nurses during their regular work shifts, without the implementation of specific handwashing protocols. The majority of samples were collected from intensive care units, outpatient clinics, and from nurses who provided post-discharge home based wound care for patients with diabetic foot or pressure ulcers.

Due to the routine submission of clinical isolates for antifungal testing by most participating hospitals, the number of environmental samples was relatively limited and primarily derived from centers with active infection control programs.

Isolates and identification

Samples were transported to the laboratory in sterile containers that contained swabs moistened with a small amount of sterile normal saline. They were subsequently inoculated onto Sabouraud dextrose agar (SDA, Condalab, Spain), which was supplemented with chloramphenicol, ciprofloxacin, and gentamycin. The cultures were incubated at 30 °C for 72 h in the dark. The culture media were checked daily to monitor for growth. For further species identification, positive cultures were sub cultured on a chromogenic medium called CHROMagar Candida (bioMérieux, Marcy l'Étoile, France), and cornmeal agar (CMA) supplemented with 1% Tween 80. This was done to assess the colony color and morphological characteristics, respectively. A distinctive cream colored colony with a mauve tinge on the chromogenic agar, along with the observation of blastoconidia along curved pseudohyphae, and giant mycelial cells in the CMA, were used to support preliminary species differentiation [11]. PCR restriction fragment length polymorphism (RFLP) analysis was performed using the MspI restriction enzyme for confirmation. Genomic DNA was extracted, and amplification of DNA was carried out using universal primers ITS1 and ITS4, as previously described [12]. In summary, 10 µl of PCR product was digested by 1.5 µl of digestion buffer, 5 units of *MspI* enzyme (Takapouzist, Iran), and distilled water to achieve a final volume of 15 µl, followed by incubation at 37 °C for 2 h. The resulting restriction fragments were visualized on a 2% agarose gel [13]. The restriction enzyme was fail to digest the product from the *C. parapsilosis* complex isolates, indicating the absence of its recognition site within the amplified region [14]. *C. parapsilosis* ATCC 22,019 was used as a reference strain for quality control during species identification. Additionally, the hyphal wall protein 1 (*HWP1*) gene was amplified to differentiate between members of the *C. parapsilosis complex*, including *C. parapsilosis* and *C. orthopsilosis*, as previously described. The *HWP1* gene was amplified by PCR using species specific primers for *C. parapsilosis* (Forward: CGAGGTGAATATGATGCT TGTA; Reverse: CCAACAGAATTGCTTAATACCATA) and *C. orthopsilosis* (Forward: ACCACCACCTAGTTC TGAG; Reverse: TCACTTGGAAGATTGGAATAACA) [15].

Antifungal susceptibility testing

A total of 527 confirmed *C. parapsilosis* complex isolates were evaluated for antifungal susceptibility using the broth microdilution method, in accordance with the Clinical and Laboratory Standards Institute (CLSI) M27-4th edition guidelines [16]. All antifungal drugs were dissolved in dimethyl sulfoxide (DMSO; Sigma) and subsequently diluted in RPMI buffered with 0.165 M morpholinepropanesulfonic acid (MOPS) at pH 7.0, without bicarbonate. The final concentrations prepared were as follow: 0.016 to 16 µg/ml for amphotericin B (AmB; Bristol-Myers-Squib), itraconazole (ITZ; Janssen), voriconazole (VRZ), clotrimazole (CLO; Pfizer), and miconazole (MZ; Sigma-Aldrich), 0.063 to 64 µg/ml for fluconazole (FLZ; Sigma-Aldrich), and 0.008 to 16 µg/ml for anidulafungin (AFG; Merck Sharp & Dohme BV) and caspofungin (CAS; Merck Sharp & Dohme BV). Microdilution

Table 1 Provide a summary of the nucleotide sequences for the utilized primers

Oligonucleotide	Target gene	Purpose	Sequence (5′ to 3′)	Product Size (bp)
FKS1 Outer Forward	*FKS1*	Amplification of the entire *FKS1* gene	TCTGTTGTTGGATTCTTCATTG	2670
FKS1 Outer Reverse	*FKS1*	Amplification of the entire *FKS1* gene	CACCCATATAAATTGACGAGTC	2670
FKS1 Inner Forward	*FKS1*	Sequencing the Hotspot 2 region	TGAATGCCATGATGAGAGG	681
FKS1 Inner Reverse	*FKS1*	Sequencing of the Hotspot 1 region	TTCTGATGGAACTTGATGGT	654
ERG11 Forward	*ERG11*	Amplification of the *ERG11* gene and sequencing	TTATTGGATTCCCTGGGTTG	650
ERG11 Reverse	*ERG11*	Amplification of the *ERG11* gene	TCATCAATGTCACCCGTCTC	650

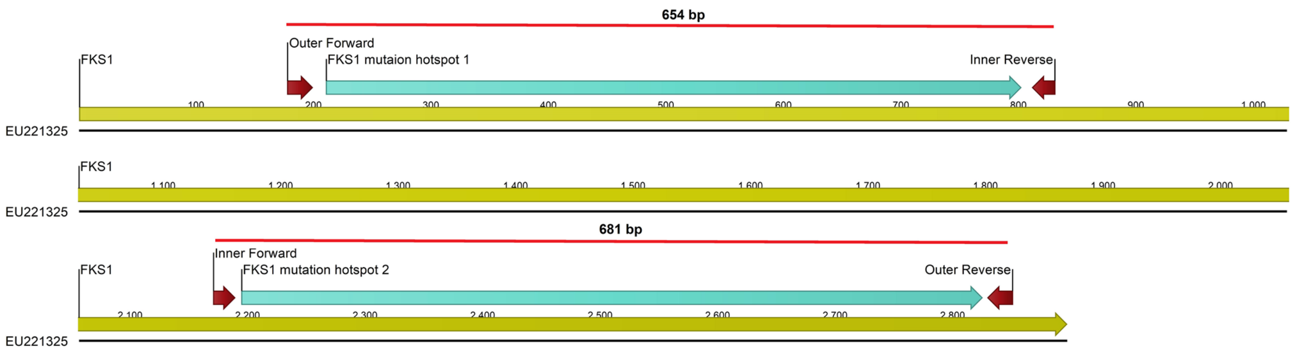

Fig. 1 Schematic map of the *FKS1* gene open reading frame of *C. parapsilosis*. The sequences shown here were originated from the NCBI GenBank database (accession no.EU221325.1)

plates were incubated at 35 °C for 24 h in the dark. and minimum inhibitory concentration (MIC) values were determined visually, defined as the lowest concentration that resulted in 50% inhibition of growth compared to the controls for all antifungals, other than amphotericin B, where the MIC was determined at 100% growth inhibition.

Antifungal agents with established clinical breakpoints (CBP) for *C. parapsilosis*, such as fluconazole, voriconazole, and echinocandins, were classified as susceptible, intermediate, or resistant according to CLSI M60 (2020) guidelines [17]. For drugs that lack of established CBP, including amphotericin B, itraconazole, miconazole, and clotrimazole, isolates were categorized as wild type (WT) or non-wild type (non-WT) based on epidemiological cutoff values (ECV) reported in CLSI M59 (2020) [18]. The MIC values of each isolate were compared with the corresponding CBP or ECV to determine their susceptibility profile.*C. parapsilosis* ATCC 22,019 was included as quality control reference on each day of testing.

Design of specific primers and PCR amplification strategy for the *FKS1* gene

Primers targeting the *FKS1* gene in echinocandin-resistant isolates were designed using AlleleID software (Premier Biosoft, USA). These primers were based on sequences derived from conserved regions identified through the NCBI database analysis (GenBank accession no. EU221325.1**)**. Before synthesis, the designed primers were evaluated for specificity, melting temperature, and potential secondary structures (Table 1). The primer design resulted in two specific sets for the *FKS1*: outer primers that produce a 2600 bp amplicon covering the entire gene, and nested inner primers that target the HS1 and HS2 regions for sequencing (Table 1; Fig. 1).

PCR amplification was performed in 20 µL reactions that contained: 10 µL Master Mix, 0.5 µL each of the outer forward/reverse primers, 2 µL DNA template (~ 50 ng), and 7 µL dH_2O. Thermal cycling conditions were as follows: 95 °C for 2 min (initial denaturation); 37 cycles of 94 °C for 30 s, 58 °C for 30 s, and 72 °C for 3 min ; followed by a final extension at 72 °C for 10 min. The success of the amplification was confirmed through agarose gel electrophoresis (Fig. 2).

Amplification of the *ERG11* gene in fluconazole resistant isolates

Primers targeting the *ERG11* gene in fluconazole-resistant isolates were designed using AlleleID software (Premier Biosoft, USA). The sequences were derived from conserved regions identified through analysis of the NCBI database, ensuring their effectiveness in detecting mutations associated with azole resistance. The reference *ERG11* gene sequence (GenBank accession no. ON209433.1) was obtained from the NCBI database and confirmed through BLAST analysis to ensure specificity for *C. parapsilosis* (Table 1).

The amplification of the *ERG11* gene was carried out in a 25 µL reaction volume, which included the following components:12.5 µL of master mix,1 µL of each forward and reverse primer (final concentration of 1 µM), 2 µL of DNA template (approximately 50 ng), and 8.5 µL

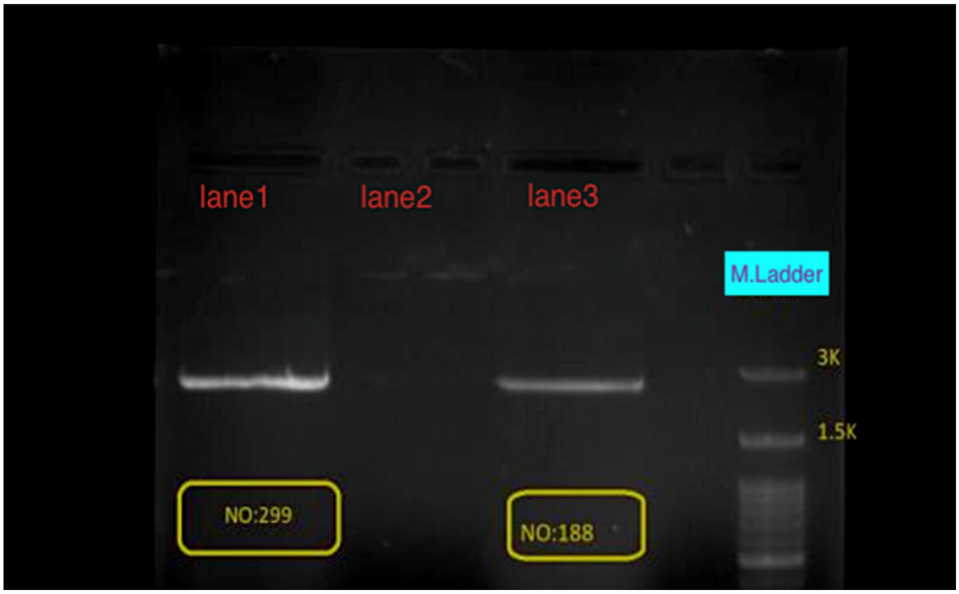

Fig. 2 Gel electrophoresis used to verify the size and integrity of the PCR products (about 2600 bp) amplifying the *FKS1* gene. Lane 1; isolate No 299, Lane 2; Negative control, Lane 3; isolate No188, M.L; Molecular ladder; 3Kb

of deionized water (dH_2O). The amplification conditions were as follows: an Initial denaturation at 94 °C for 5 min, followed by 35 cycles consisting of denaturation at 94 °C for 15 s, annealing at 56 °C for 30 s, extension at 72 °C for 30 s, and a final extension at 72 °C for 10 min. The PCR product was analyzed by electrophoresis on a 1.5% agarose gel stained with a DNA-safe dye. A single distinct band of approximately 650 base pairs was observed, corresponding to the expected size of the *ERG11* gene fragment.

Sequencing and analysis

After electrophoresis, target DNA fragments were excised and purified using the Favorgen Gel Extraction Kit (Favorgen Biotech Corp., Taiwan) following the manufacturer's protocol. All amplicons were sequenced using Sanger sequencing, and subsequent sequence analyses were performed using MEGA 11 software to identify potential mutations in the targeted regions.

Results

A total of 2,600 samples were collected during the study period from patients, healthcare workers' hands, and environmental surfaces. From these, 527 isolates were identified at the species level using PCR amplification of the ITS region followed by RFLP analysis. This approach is a well-established and reliable method for Candida species identification, enabling differentiation based on characteristic restriction fragment patterns. To further confirm the accuracy of species identification within the *C. parapsilosis* complex, amplification of the *HWP1*gene was also performed. This amplification produced PCR products of approximately 840 bp for *C. parapsilosis* and 900 bp for *C. orthopsilosis*. All studied yeast isolates showed an expected band and was in "*psilosis*" clade.

Out of 527 *C. parapsilosis* complex isolates, 510 Clinical samples were obtained from various body sites. Among the clinical samples, the majority were derived from burn wounds (165 isolates, 32.4%), followed by onychomycosis (84 isolates, 16.5%), neonatal skin (47 isolates, 9.2%), diabetic foot ulcers (30 isolates, 5.9%), axillary regions (26 isolates, 5.1%), groin (24 isolates, 4.7%), sputum (23 isolates, 4.5%), blood (22 isolates, 4.3%), vaginal swabs (16 isolates, 3.1%), ear secretions (15 isolates, 2.9%), and bronchoalveolar lavage (BAL) samples (15 isolates, 2.9%). Other sources included mouth swabs (9 isolates, 1.8%), toe/finger web spaces (8 isolates, 1.6%), eye secretions and tears (6 isolates, 1.2%), decubitus ulcers (5 isolates, 1.0%), neck samples (5 isolates, 1.0%), nasal swabs (4 isolates, 0.8%), urine (3 isolates, 0.6%), the sole of the foot (2 isolates, 0.4%), and the scalp (1 isolate, 0.2%).These samples were collected from multiple cities across Iran, such as Mashhad, Dezful, Sari, Rasht, Qaemshahr, and Tehran. Furthermore, 11 samples from HCWs were obtained through composite hand swabs from nurses working in intensive care units, outpatient clinics, and nursing home care units. Six environmental samples were collected from reusable breast pump sets used in a hospital setting. A summary of the isolation sources and sampling contexts is provided in Table 2.

Yeast colonization on hcws' hands

A total of ninety samples were collected from the hands of 90 nurses. Among these, 52 nurses worked in intensive

Table 2 Origin and distribution of *C. parapsilosis* complex isolates from clinical, environmental, and healthcare worker sources

Origin of the sample	Sample Type	Source of positive isolates: *C. parapsilosis* complex (*n* = 527)	% of Total Isolates	Collection details
Clinical *n* = 2180	**Various Swabs (n =** 428) includes samples from the following sites: axilla, groin, neck, scalp, vagina, ear, eye (tears/secretions), oral cavity, foot sole, interdigital spaces (hands/feet), and decubitus ulcer **Respiratory Samples** (*n* = 208) includes: bronchoalveolar lavage (BAL) samples (70), sputum (48), Nasal swabs (90) burn sites biopsies (*n* = 697) nails (*n* = 380) blood samples (*n* = 92) scalp (*n* = 35) neonatal skin swab (*n* = 190) diabetic foot ulcers (*n* = 150)	***n* = 510** burn wounds (165 isolates, 32.4%), onychomycosis (84 isolates, 16.5%), neonatal skin (47 isolates, 9.2%), diabetic foot ulcers (30 isolates, 5.9%), axillary regions (26 isolates, 5.1%), groin (24 isolates, 4.7%), sputum (23 isolates, 4.5%), blood (22 isolates, 4.3%), vaginal swabs (16 isolates, 3.1%), ear secretions (15 isolates, 2.9%), bronchoalveolar lavage (BAL) samples (15 isolates,2.9%), mouth swabs (9 isolates, 1.8%), toe/finger web spaces (8 isolates, 1.6%), eye secretions and tears (6 isolates, 1.2%), decubitus ulcers (5 isolates, 1.0%), neck samples (5 isolates, 1.0%), nasal swabs (4 isolates, 0.8%), urine (3 isolates, 0.6%), the sole of the foot (2 isolates, 0.4%), scalp (1 isolate, 0.2%)	96.8%	Collected from patients admitted in hospitals across Iran
Environmental *n* = 330	Inanimate surfaces such as bed rails (78), cardiac monitors (60), ventilators (46), infusion pumps (28), dialysis devices (52), sinks (13), cabinets near patient beds (20), doorknobs (21), reusable breast pump sets (12)	***n* = 6** reusable breast pump sets (100%)	1.1%	Environmental samples were collected from hospitals affiliated with Mazandaran University of Medical Sciences.
Healthcare workers' hands *n* = 90	hand swabs from nurses	***n* = 11** 6 were obtained from ICU/NICU staff, 4 from outpatient clinic nurses, and 1 from a home care nurse	2.1%	Collected from healthcare workers in hospitals affiliated with Mazandaran University of Medical Sciences during regular work shifts, prior hand hygiene

Table 3 Prevalence of yeast colonization among healthcare workers based on unit and gender

Group	Total Nurses (*N*)	Nurses with Yeast Colonization (*n*)	Percentage of Yeast Colonization	Male Nurses Positive (*n*)	Female Nurses Positive (*n*)	*C. parapsilosis* Isolates
ICU, NICU* Nurses	52	37	71%	16	21	6
Outpatient Clinic Nurses	35	28	80%	10	18	4
HCNs **	3	3	100%	3	0	1
Total (All Nurses)	90	68	75%	29	39	**11**

*NICU; Neonatal Intensive Care Unit. ** HCNs: Home care nurses

care units, 35 worked in outpatient clinics, and 3 were home care nurses. The sampling occurred during their normal work shifts, without any specific handwashing or preparatory measures. It was found that 75% (*n* = 68) of the nurses carried yeasts on their hands. This included 71% (*n* = 37) of the intensive care unit nurses, 80% (*n* = 28) of the outpatient clinic nurses, and all 3 (100%) of the home care nurses. Within the hand cultures tested, 29 out of 36 male nurses (83%) and 39 out of 54 female nurses (73%) were positive for yeast colonization. A total of 11 *C. parapsilosis* isolates were obtained from the collected samples (see Table 3). No isolates of *Candida orthopsilosis* or *Candida metapsilosis* were found. Among the isolates recovered from healthcare workers' hands, *C. parapsilosis* complex was the second most frequently isolated species after the *Candida albicans* complex.

Antifungal susceptibility profile

Antifungal susceptibility testing was performed according to CLSI) M27-4th edition guidelines (Table 4). Among the 527 isolates, which included clinical, environmental, and HWC hand-derived samples, 10 isolates

Table 4 In vitro antifungal susceptibility of eight antifungal agents against total *C. parapsilosis* complex isolates sourced from clinical settings, environmental samples, and healthcare workers in Iran

Sample type of *C. parapsilosis*	Type of complex of *C. parapsilosis*	Anti-fungal	No. of isolates / MIC (µg/mL)													(µg/mL)			
			64	**32**	**16**	**8**	**4**	**2**	**1**	**0.5**	**0.25**	**0.125**	**0.063**	**0.032**	**0.016**	**MIC Range**	**MIC 50**	**MIC 90**	**G mean**
Clinical($n = 510$)	*C. parapsilosis sensu stricto* ($n = 507$)	FLZ	0	0	2	0	**150**	86	88	65	59	41	16	0	0	0.063-16	1	4	1.02
		VRZ	0	0	1	0	0	0	0	65	71	8	62	78	**142**	0.016-16	0.063	0.5	0.07
		CLO	0	0	0	0	0	0	0	55	51	88	83	62	**168**	0.016-0.5	0.063	0.5	0.06
		MZ	0	0	0	0	0	2	54	84	83	**112**	83	44	45	0.016–2	0.125	1	0.147
		ITZ	0	0	0	1	0	0	0	107	102	**125**	72	69	33	0.016-8	0.125	0.5	0.127
		CAS	0	0	8	0	59	53	49	60	50	38	64	**75**	51	0.016-16	0.25	4	0.252
		AFG	0	0	1	2	18	40	41	31	40	35	89	94	**114**	0.016-16	0.063	2	0.106
		AMB	0	0	0	0	0	0	39	108	**149**	130	65	8	8	0.016-1	0.25	0.5	0.209
	C. orthopsilosis ($n = 3$)	FLZ	0	0	0	0	0	0	0	1	**0**	2	0	0	0	0.125-0.5	0.125	0.5	0.25
		VRZ	0	0	0	0	0	0	0	0	**2**	1	0	0	0	0.125–0.25	0.25	0.25	0.19
		CLO	0	0	0	0	0	0	0	0	**2**	1	0	0	0	0.125–0.25	0.25	0.25	0.198
		MZ	0	0	0	0	0	0	0	2	**1**	0	0	0	0	0.25–0.5	0.5	0.5	0.39
		ITZ	0	0	0	0	0	0	0	2	**0**	0	1	0	0	0.063-0.5	0.5	0.5	0.25
		CAS	0	0	0	0	0	0	0	0	**2**	0	0	1	0	0.032–0.25	0.25	0.25	0.126
		AFG	0	0	0	0	0	0	0	0	**0**	2	1	0	0	0.063–0.125	0.125	0.125	0.1
		AMB	0	0	0	0	0	0	0	0	**1**	1	1	0	0	0.063–0.25	0.125	0.25	0.126
Environmental ($n = 6$)	*C. parapsilosis sensu stricto* ($n = 6$)	FLZ	0	0	0	0	1	0	1	**2**	0	2	0	0	0	0.125-4	0.5	2.5	0.5
		VRZ	0	0	0	0	0	0	0	0	0	0	**3**	0	3	0.016–0.063	0.039	0.063	0.031
		CLO	0	0	0	0	0	0	0	1	0	0	1	0	**4**	0.016-0.5	0.016	0.28	0.035
		MZ	0	0	0	0	0	0	0	1	1	**2**	0	0	2	0.016-0.5	0.125	0.375	0.089
		ITZ	0	0	0	0	0	0	0	1	**3**	0	1	1	0	0.032-0.5	0.25	0.375	0.158
		CAS	0	0	0	0	0	1	**3**	1	1	0	0	0	0	0.25-2	1	1.5	0.793
		AFG	0	0	0	0	0	0	0	**2**	2	2	0	0	0	0.125-0.5	0.25	0.5	0.25
		AMB	0	0	0	0	0	0	1	0	**5**	0	0	0	0	0.25-1	0.25	0.625	0.315
HWCs'hand ($n = 11$)	*C. parapsilosis sensu stricto* ($n = 11$)	FLZ	0	0	0	0	0	0	0	1	**6**	4	0	0	0	0.125-0.5	0.25	0.45	0.206
		VRZ	0	0	0	0	0	0	0	0	3	1	1	1	**5**	0.016–0.25	0.032	0.25	0.049
		CLO	0	0	0	0	0	0	0	1	1	0	2	1	**6**	0.016-0.5	0.016	0.45	0.038
		MZ	0	0	0	0	0	0	3	**6**	1	0	0	1	0	0.032-1	0.5	1	0.441
		ITZ	0	0	0	0	0	0	0	0	0	0	2	**9**	0	0.032–0.063	0.032	0.063	0.036
		CAS	0	0	0	0	0	0	0	0	**9**	2	0	0	0	0.125–0.25	0.25	0.25	0.22
		AFG	0	0	0	0	0	0	0	0	3	1	3	1	**3**	0.016–0.25	0.063	0.25	0.063
		AMB	0	0	0	0	0	0	0	0	0	**7**	4	0	0	0.063–0.125	0.125	0.125	0.097

Table 4 (continued)

Sample type of *C. parapsilosis*	Type of complex of *C. parapsilosis*	Anti-fungal	No. of isolates / MIC (μg/mL)													(μg/mL)			
			64	**32**	**16**	**8**	**4**	**2**	**1**	**0.5**	**0.25**	**0.125**	**0.063**	**0.032**	**0.016**	**MIC Range**	**MIC 50**	**MIC 90**	**G mean**
Total(*n* = 527)	Total *C. parapsilosis* complex	FIZ	0	0	2	0	**151**	86	89	69	65	49	16	0	0	0.063-16	1	4	0.983
		VRZ	0	0	1	0	0	0	0	65	76	90	66	79	**150**	0.016-16	0.063	0.5	0.069
		CLO	0	0	0	0	0	0	0	57	54	89	86	63	**178**	0.016-0.5	0.063	0.5	0.059
		MZ	0	0	0	0	0	2	57	93	86	**114**	83	45	47	0.016-2	0.125	1	0.15
		ITZ	0	0	0	1	0	0	0	108	105	**125**	76	79	33	0.016-8	0.125	0.5	0.125
		CAS	0	0	8	0	59	54	52	61	62	40	64	**76**	51	0.016-16	0.25	4	0.255
		AFG	0	0	1	2	18	40	41	33	45	42	93	95	**117**	0.016-16	0.063	2	0.106
		AMB	0	0	0	0	0	0	40	108	**155**	138	70	8	8	0.016-1	0.25	0.5	0.206

FLZ, Fluconazole; VRZ, Voriconazole; ITZ, Itraconazole; AFG, Anidulafungin; and AMB, Amphotericin B. CAS, caspofungin; MZ, Miconazole; CLO, Clotrimazole; MIC, Minimum inhibitory concentration. GM, Geometric mean value; MIC50/90, MIC required to inhibit 50% and 90% of the isolates, respectively

(1.897%) showed in vitro resistance based on CLSI breakpoints. Specifically, two isolates (0.38%) exhibited resistance to fluconazole, five isolates (0.95%) were resistant to caspofungin, and three isolates (0.57%) exhibited resistance to both caspofungin and anidulafungin.

Fluconazole showed a relatively wide MIC range of 0.063–16 μg/mL, with MIC50 and MIC90 values of 1 and 4 μg/mL, respectively. Although most isolates fell within the susceptible dose-dependent range, a small proportion, particularly among clinical isolates, showed elevated MIC indicative of reduced susceptibility. Voriconazole (VRZ) demonstrated strong activity, with an overall MIC range of 0.016–16 μg /mL. The MIC50 and MIC90 values of 0.063 and 0.5 μg/mL resulting in a geometric mean (Gmean) MIC of voriconazole was 0.069 μg /mL, indicating potent effectiveness, especially among non clinical isolates. Clotrimazole (CLO) also exhibited strong in vitro activity, with a MIC50 (0.063 μg /mL), a MIC90 (0.5 μg /mL), and a low Gmean MIC (0.059 μg /mL).

No resistant isolates were found in any of the groups tested. Miconazole (MZ) showed moderate activity, with a broader MIC range of 0.016–2 μg /mL. The MIC50 and MIC90 values were 0.125 μg /mL and 1 μg /mL, respectively, resulting in a Gmean MIC (0.15 μg /mL). This indicates a relatively higher MIC among clinical isolates compared to environmental and hand derived isolates from HWC. Itraconazole (ITZ) showed a narrow MIC range of 0.016–8 μg /mL, with MIC 50 and MIC 90 values of 0.125 and 0.5 μg /mL, respectively. The Gmean MIC was 0.125 μg /mL, reflecting consistent susceptibility across all 527 sources of isolates.

Among the echinocandins, caspofungin (CAS) showed reduced activity, with a wide MIC range of 0.016–16 μg /mL, and an elevated MIC 90 of 4 μg /mL, suggesting intrinsic reduced susceptibility in the *C. parapsilosis* complex. The Gmean MIC for caspofungin was 0.255 μg /mL. Anidulafungin (AFG) demonstrated potent activity with MIC50 and MIC90 values of 0.063 and 2 μg / mL, respectively, leading to a Gmean MIC of 0.106 μg / mL. While most isolates were susceptible, some clinical isolates showed higher MICs.Amphotericin B (AMB) showed consistent activity, with a MIC range of 0.016 to 1 μg /mL, MIC50 and MIC90 values of 0.25 and 0.5 μg /mL, respectively. The Gmean MIC was 0.206 μg /mL, with no evidence of resistance. Among the tested drugs, voriconazole, clotrimazole, and amphotericin B demonstrated potent activity against the majority of isolates, with overall low MIC90 values of 0.5 μg/mL. In contrast, caspofungin showed less in vitro activity.

Molecular analysis of the resistant isolates

Ten isolates (1.897%) exhibited in vitro resistance according to CLSI breakpoints. Specifically, two isolates (0.38%) were resistant to fluconazole, while eight isolates (1.52%)

demonstrated resistance to echinocandins. Using newly designed outer and inner primers that target the *FKS1* gene, the hotspot regions (HS1 and HS2) were successfully amplified and sequenced in the majority of *C. parapsilosis* complex isolates. This facilitated the accurate detection of potential resistance associated mutations. To investigate the molecular basis of antifungal resistance, the *ERG11* and *FKS1* genes in the resistant isolates were sequenced; however, no mutations were found in the sequenced regions of both genes. The obtained sequences have been deposited in the NCBI GenBank database with the accession numbers PV764592 to PV764597 and PQ721052.

Discussion

C. parapsilosis is one of the most clinically relevant non-*albicans Candida* (NAC) species, particularly in hospital setting. It is globally recognized as the second or third most frequently isolated *Candida* species in intensive care unit (ICU) [19]. In this study, we analyzed the antifungal susceptibility profiles and resistance associated genetic markers in 527 isolates of the *C. parapsilosis* complex. Similar to *Candida auris*, *C. parapsilosis* frequently demonstrates multidrug resistance and possesses a high potential for hospital transmission through direct patient contact [20].

In this study, clinical isolates showed higher MIC for fluconazole and echinocandins compared to isolate from environmental sources and healthcare workers' hands. Similar to a previous study conducted in 2021, the researcher observed greater resistance to fluconazole in clinical isolates [21]. In our study, fluconazole-resistant strains of *C. parapsilosis* were exclusively found in clinical samples, particularly from sites of active infection, with no resistance detected on hands or in environmental sources. This pattern likely indicates increased antifungal pressure in clinical settings, as evidenced by elevated MIC90 values for fluconazole and caspofungin among clinical isolates. In contrast, all environmental isolate and those from healthcare workers remained fully susceptible to all tested antifungal agents, suggesting limited antifungal exposure and low selective pressure in non-clinical settings. Additionally, none of the *C. parapsilosis* complex isolates in our study showed resistance to amphotericin B, confirming the drug's continued efficacy against this species in our setting.

This finding aligns with a review article [19] that assessed 28 studies, where six reported no resistance to amphotericin B, based on CLSI guidelines and/or E-test (gradient diffusion assay).

The study highlights that voriconazole (GM = 0.0689) and clotrimazole (GM = 0.059) exhibit strong activity against *C. parapsilosis*, suggesting that voriconazole is more effective than fluconazole for treating candida infections [22]. Our data indicate that clotrimazole (CLO) is highly effective across all *C. parapsilosis* isolates. Similarly, previous research in 2015 found that all isolates from onychomycosis patients were susceptible to CLO with a geometric mean of 0.35 µg/ml. Both studies highlight clotrimazole as a valuable treatment option for *Candida* infections, particularly in cases resistant to other azoles, such as FLZ [23]. In line with a study conducted in 2020 [24], our findings showed high in vitro susceptibility of clinical isolates within the *C. parapsilosis* species complex to itraconazole with a MIC90 of 0.5 µg/mL. In a study examining 101 Iranian isolates, 89.1% were found to be wild type to itraconazole. This supports the continued efficacy of itraconazole against *C. parapsilosis* and highlights the importance of species level identification. In contrast, Hassanmoghadam et al. [25] reported that 89% (n = 94) of *C. parapsilosis* isolates from the various body sites had MIC values ≥ 1 µg/mL, indicating resistance. This discrepancy may reflect regional differences or prolonged use of itraconazole for prophylaxis or treatment within their patient population. In a study conducted in Mexico [26] all clinical isolates of *C. parapsilosis* were found to be susceptible to FLZ, amphotericin B, 5-fluorocytosine, and ketoconazole, while showing resistance to miconazole and itraconazole. In contrast, our study found no resistance to itraconazole, suggesting potential regional differences or variations in antifungal usage. Our findings align with a study by Kord et al. [27] in Iran, which reported high MIC values for echinocandins against clinical isolates of *C. parapsilosis* complex. Specifically, the MIC50 and MIC90 values were 1 µg/mL and 4 µg/mL for caspofungin, and 2 µg/mL and 4 µg/mL for anidulafungin, respectively. These values were notably higher compared to other non-*albicans Candida* species. Similarly, our study showed elevated MIC90 for these drugs (4 µg/mL for caspofungin and 2 µg/mL for anidulafungin), raising concerns about reduced susceptibility .In contrast, another study [9] found that all tested isolates of *C. parapsilosis* showed full susceptibility to micafungin and caspofungin (MIC < 4 µg/mL). A Brazilian study [28] involving 124 pediatric clinical isolates reported no resistance profile, which may be attributed to geographic, demographic, antifungal prescribing, or local epidemiological factors that influence the emergence of resistance.

Global surveillance study revealed outbreaks of FLZ resistant *C. parapsilosis*, often linked to the nosocomial transmission of genetically related strains [29–31]. However, in our study, only two FLZ resistant clinical isolates were identified, resulting in a prevalence of 1.96% among the 510 isolates, which aligns with global resistance estimates (2% to 5%) [1]. This highlights the necessity of routine antifungal susceptibility testing, especially since FLZ continue to be the standard first line treatment, particularly in resource limited settings. Furthermore, FLZ prophylaxis and cross-transmission within healthcare settings may foster the emergence and spread of azole-resistant *C. parapsilosis* strains [32, 33].

Recently, a rise in *C. parapsilosis* candidemia was observed during the COVID-19 pandemic at a large hospital in Seoul [34], but it was not linked to a specific clone (Y132F), due to the low rate of FLZ resistance [35]. Comprehensive antifungal stewardship programs are highly recommended in choosing antifungal treatment regimens and mitigating resistance patterns [36]. Moreover, incorporating fungal biomarker testing into ICU protocols can help reduce unnecessary FLZ use by enabling targeted empirical therapy for high risk patients [37]. Persistent environmental contamination by *C. parapsilosis* and C. *auris* in healthcare settings continues to pose a significant risk for hospital acquired infections [1]. Several studies linked outbreaks of *C. parapsilosis* to biofilm on Central Venous Catheters)CVCs (and hand transmission through healthcare workers' hands, emphasizing the need for rigorous surveillance and strict infection control measures [38–42]. Our study identified six *C. parapsilosis* isolates on equipment in a neonatal unit, specifically on breast pump sets, indicating a risk of infection from contaminated devices.

Notably, a multicenter study reported a 46.6% in hospital mortality rate among patients with *C. parapsilosis* complex candidemia in Iran from 2015 to 2019 [43]. Effective infection control is crucial, especially for vulnerable groups like neonates and patients receiving parenteral nutrition, who are at a higher risk of *C. parapsilosis* candidemia due to biofilm formation on CVCs and transmission from healthcare workers [1]. In a related study, an outbreak of *C. parapsilosis* candidemia in pediatric oncology patients was linked to bloodstream isolates detected on the hands of healthcare workers, underscoring the role of inadequate infection control practices, including improper PPE use and contaminated hospital environments, in facilitating nosocomial transmission [44]. Both an earlier outbreak [45] and the recent outbreak [46] reported inadequate PPE use, which led to the contamination of healthcare workers' hands and shared computer keyboards. This highlights the role of improper PPE practices in nosocomial transmission of infections. Consequently, both contaminated surfaces and healthcare workers' hands pose significant risks to patients and staff [47].

A significant percentage of *C. parapsilosis* was detected on hands of healthcare workers, specifically (57%) [48], which is comparable to the 59.7% found in clinical specimens. Identical genotypes of *C. parapsilosis* present in both hand and blood samples suggest a possibility of horizontal transmission. Our study identified *C. parapsilosis* in 11 healthcare workers, emphasizing the urgent need for reinforced infection control measures and strict adherence to hand hygiene, especially in high-risk areas such as the NICU.

In addition, three isolates were identified as *C. orthopsilosis*, all of which were collected from patients. None of these isolates showed resistance to the tested antifungal agents. Due to their limited number, they were not analyzed individually; however, their MIC values fell within the susceptible range. This finding aligns with previous studies indicating that *C. orthopsilosis* typically exhibits lower MIC values and has limited clinical significance compared to *C. parapsilosis* sensu stricto [49, 50].

Our analysis of resistant *C. parapsilosis* isolates showed no mutations in the *ERG11* or *FKS1* hotspot regions. We amplified and sequenced HS1 and HS2 regions of the *FKS1* gene, consistent with previous studies [9, 43], which also found no mutations associated with echinocandin resistance. Sequencing of the *ERG11* gene revealed no FLZ resistance mutations, including the Y132F mutation. This aligns with earlier findings that observed reduced azole susceptibility in the absence of *ERG11* mutations [43]. Our results contrast with earlier studies [21] that linked *ERG11* mutations to azole resistance. The absence of these mutations in our isolates suggests alternative mechanisms of resistance, such as overexpression of efflux pump genes (e.g., CDR1, MDR1), modifications in membrane sterol composition, or novel mutations outside the traditional hotspots. Additionally, factors such as biofilm formation and stress response pathways (e.g., heat shock proteins, oxidative stress regulators) may also contribute to the observed reduced susceptibility, as noted in prior investigations [51, 52]. Further molecular analyses, such as transcriptomics or whole genome sequencing (WGS), are needed to clarify the resistance mechanisms. This highlights the ongoing need for surveillance, rapid diagnostics, and antifungal stewardship. A 2010 study published in the European Journal of Pediatrics linked an outbreak of *C. parapsilosis* in pediatric patients to the hands of healthcare workers through molecular analyses [26].

In 2021, a comprehensive environmental screening [21], consistent with the previous findings [45, 53], revealed that *C. parapsilosis* was the most prevalent *Candida* species found on HCWs' hands and surfaces. This finding indicates a potential for the horizontal transmission of a specific clone. However, a study conducted by Sharma et al. (2025) in North India found no dominant clone among FLZ resistant *C. parapsilosis* isolates from various clinical specimens. This suggests that resistance can arise from multiple independent introductions rather than from a single source [54].

The practical validation of the designed primers is crucial. In our study, we tested newly designed primers targeting the *FKS1* gene, including the hotspot regions (HS1 and HS2) on clinical isolates of the *C. parapsilosis* complex to ensure their specificity and reliability. We also optimized the PCR conditions, such as primer concentrations, melting temperatures, and annealing temperatures to maximize amplification efficiency and minimize nonspecific products. These steps are essential to confirm that the primers are suitable for downstream sequencing and potential routine diagnostic applications. To enhance workflow efficiency and reduce experimental costs, e designed a single pair of primers that amplifies a PCR fragment containing both hotspot 1 and hotspot 2 regions of the *FKS1* gene. This approach significantly decreases the number of required PCR reactions and sequencing runs, leading to lower reagent consumption, shorter turnaround times, and overall cost savings. Furthermore, it enables the simultaneous assessment of both mutation-prone regions in a single amplification run, thereby improving data consistency and facilitating direct comparison of mutation patterns. This approach offers a more efficient and cost-effective alternative to amplifying the two hotspots separately. However, this study has several notable limitations. Firstly, we were unable to perform molecular typing techniques, which could have provided a clearer understanding of clonal relationships and transmission dynamics in healthcare-associated infections. Future studies should incorporate genotyping strategies to enhance epidemiological insights into the transmission of *C. parapsilosis*. Secondly, while we sequenced the *ERG11* and *FKS1* hotspots to investigate antifungal resistance, we did not examine other resistance-associated genes or perform gene expression analyses. These additional analyses could provide valuable information, especially for FLZ resistant strains lacking known mutations. Furthermore, only six environmental *C. parapsilosis* complex isolates were recovered in this study, which limits the generalizability to our conclusions about environmental contamination and its role in hospital transmission. Lastly, the absence of clinical data from patients with the isolates hindered our ability to assess treatment outcomes or draw correlations. To address these issues, coordinated surveillance systems at local, national, and international levels are essential for monitoring emerging resistance patterns. Integrating WGS into these efforts will help to track genotype circulation and accurately identify transmission networks, ultimately enabling better management within institutions to inform therapeutic decision-making.

Conclusion

This study highlights the antifungal susceptibility patterns of *C. parapsilosis* complex isolates from clinical, environmental, and healthcare worker sources in Iran. Overall, a low prevalence of antifungal resistance was observed, with newer triazoles showing consistent in vitro activity. No resistance associated mutations were detected in the *ERG11*or *FKS1* genes, suggesting that alternative resistance mechanisms may be involved. These findings underscore the importance of continued surveillance and molecular investigations to guide effective antifungal therapy and infection control strategies. However, further studies are warranted to correlate these findings with clinical outcomes.

Abbreviations

CLSI	Clinical Laboratory Standards Institute
PCR	Polymerase Chain Reaction
DMSO	Dimethyl sulfoxide

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Author contributions

A.M., T.S., and H.B. made the primary contributions to writing the manuscript. M.A., M.S., R.V., M.Z., M.T.H., S.R.A., Z.A., L.F., I.H., H.Z., and S.M. oversaw project follow-up, conducted literature searches, and contributed to revising the manuscript. All authors read, reviewed, and approved the final version.

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Data availability

Data is provided within the manuscript files. The DNA sequences generated during the current study have been deposited in the NCBI GenBank repository under accession numbers PV764592 to PV764597 and PQ721052.

Declarations

Ethics approval and consent to participate

All human subjects, materials, or data procedures were conducted by ethical standards established by the institutional and/or national research committee, adhering to the 1964 Helsinki Declaration and its subsequent amendments or comparable ethical guidelines. All participants, or their parents or legal guardians, provided informed consent for their participation in the study. This study was approved by the Ethics Committee of Mazandaran University of Medical Sciences, under the approval number IR.MAZUMS.REC.1402.365. All procedures were conducted according to the ethical guidelines and regulations.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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